

Flexi-JEGNN — Architecture

SMILES Input e.g. CC(=O)Oc1ccccc1C(=O)O — no 3D coordinates required at any level

Atom Feature Extraction (RDKit — identical across all distance levels)
Atom type one-hot (12-d) | Degree | Formal charge | H-count | Aromaticity | Ring | Chirality
Features encode pharmacophoric primitives (H-bond, charge, lipophilicity)

One of five distance levels is selected at runtime — all other components are shared exactly

L0 Constant Å
All bonded pairs
same distance.
Binary adjacency.

7.7 ms/mol

L1 Topological Å
Dijkstra shortest
path bond count.
Topology, no geometry.

10.5 ms/mol

L2 — Bounds Midpoint
(Gasteiger et al. 2021)
RDKit Distance Geometry.
No conformer needed.
MAE = 0.254 Å

0.72 ms/mol ✓ 159×

L3 ETKDgv3 Truth
Explicit 3D conformer.
MMFF94 optimised.
Ground-truth ceiling.

114.7 ms/mol

L4 Random Null Å
Per-molecule seed.
Zero physical info.
Null control.

29.3 ms/mol

PARETO RESULT
L0/L1 cost zero spatial
computation yet are
statistically
indistinguishable from L3
on all 6 tasks.

Spatial Threshold Filter
Directed edge ($i \rightarrow j$) created iff $e_{ij} < \tau$, where $\tau = 5.0$ Å
(sensitivity analysis at $\tau \in \{3.0, 5.0, 8.0\}$ Å: Appendix C)

Gaussian Smearing — Radial Basis Functions
 $\text{RBF}(e_{ij})_k = \exp(-\gamma(e_{ij} - \mu_k)^2)$, $k = 0, \dots, 15$
 μ_k evenly spaced in $[0.0, 5.0]$ Å; 16 centres
One scalar $\rightarrow$ 16-dimensional continuous position fingerprint on the 0–5 Å axis

Bond type one-hot:
[single, double, triple,
aromatic, ring]

Same for all L0–L4

Joint-Edge Concatenation
 $\mathbf{E}_{ij} = [\text{RBF}(e_{ij}) \parallel \text{bond-type}] \in \mathbb{R}^{21}$
Only the first 16 dimensions change between L0–L4. The bond-type dimensions are level-invariant.

JointEdgeConv $\times 5$ (layers = 5 ; hidden dim = 256 ; receptive field = 5 hops)
Message: $\mathbf{m}_{ij}^{(l)} = \text{MLP}_{\text{msg}}(\mathbf{h}_i^{(l-1)} \parallel \mathbf{h}_j^{(l-1)} \parallel \mathbf{E}_{ij})$
Update: $\mathbf{h}_i^{(l)} = \mathbf{h}_i^{(l-1)} + \text{LN}(\text{MLP}_{\text{upd}}(\mathbf{h}_i^{(l-1)} \parallel (1/l, l(i)) \parallel \sum_{j \in \mathcal{N}(i)} \mathbf{m}_{ij}^{(l)}))$
After 5 rounds each atom embedding encodes up to 5-hop neighbourhood context
[5 layers and dim=256 validated by ablation: Appendix A]

4.2 M parameters
Hidden dim: 256
Dropout: 0.3
LN on residual
Loss: Weighted BCE
Seeds: $n = 20$

Multi-Aggregation Readout (PNA)

Sum
 $\sum_i \mathbf{h}_i^{(5)}$
Molecular extensivity
(total electron count,
molecular size)

Mean
 $(1/N) \sum_i \mathbf{h}_i^{(5)}$
Average chemical environment

Max
 $\max_i \mathbf{h}_i^{(5)}$
Dominant / most
reactive functional group

Concatenate [Sum $\parallel$ Mean $\parallel$ Max] $\rightarrow \mathbf{H}_{\text{graph}} \in \mathbb{R}^{768}$; $d = 256, 256 \times 3 = 768$

2-Layer MLP Regressor
 $\hat{y} = \mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{H}_{\text{graph}} + \mathbf{b}_1) + \mathbf{b}_2$

Regression Results
(e.g., QM9 HOMO-LUMO, PDBbind Affinity)

2-Layer MLP Classifier
 $\hat{y} = \sigma(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{H}_{\text{graph}} + \mathbf{b}_1) + \mathbf{b}_2)$

Classification Results
(e.g., BACE, ADMET Clearance/Toxicity)